

A cartographic approach to Chinese V *de* O clefts

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Abstract: This paper investigates the so-called V *de* O clefts in Chinese, a specific type of “clefts” where the *de* particle appears between the verb and the object, and proposes a cartographic approach to their syntactic and semantic properties. While some common generalizations about this pattern are shown to hold upon closer examination (i.e., exhaustive focus, obligatory past reading, ban on TMA markers), some do not (i.e., strict adjacency effect, term focus restriction). Two key projections are argued to exist in the relevant functional hierarchy, each linked to one crucial element in the pattern: (i) a FocP selected by the copula *shi*; (ii) an Asp*P headed by the verbal particle *de*. The former explains the focus effects with syntactic encoding and the latter accounts for the temporal/aspectual peculiarities with the marking of perfective aspect (related predictions about predicate restrictions are also presented). The derivation of V *de* O clefts involves overt focus movement of an extended “eventive” projection, which minimally contains an Asp*P. This movement is triggered by a focus feature either on the whole constituent (VP/S+VP) or on a smaller constituent within it (verb/object/subject/adjunct), the latter case demonstrating a pied-piping movement.

Keywords: Chinese V *de* O clefts; cartography; copula; Focus projection; perfective aspect

1. V *de* O clefts in (northern) Mandarin Chinese

The so-called V *de* O clefts as in (1) are widely agreed to be one of the Chinese equivalents of English *it*-clefts:

(1) V *de* O pattern

*Shi wo dapo de na-ge huaping*¹.

be 1SG break DE that-CL vase

‘It was I who broke that vase.’

Similar to their English counterparts, V *de* O clefts are also believed to show a focus-presupposition bipartition (Paul and Whitman 2008; Hole 2011 among others), with the focused element immediately following the copula *shi*²:

¹ The following abbreviations are used in the glosses: 1/2/3—first/second/third person; SG—singular; CL—classifier; NEG—negation; PFV—perfective aspect; SFP—sentence-final particle.

² For ease of reference, *shi* will be referred to as “the copula” throughout descriptive discussions here, despite different analyses in existing literature for the syntactic status of this element (cf. Section 2.2).

(2) Focus-presupposition bipartition

Shi [wo]_{focus} [*dapo de na-ge huaping*]_{presupposition}.
be 1SG break DE that-CL vase
'It was [I]_{focus} [who broke that vase]_{presupposition}.'

In addition to the verb-adjacent position, the functional element *de* can also occur in a sentence-final position, resulting in another cleft pattern³:

(3) V O *de* pattern

Shi wo dapo na-ge huaping de.
be 1SG break that-CL vase DE
'It was I who broke that vase.'

Despite the apparent free alternation⁴, the above two patterns do show differences in terms of distribution and interpretation (see Paul and Whitman 2008 for an overview; also cf. Section 5). One distinctive property of V *de* O clefts concerns tense/time: a past time reading of the sentence is obligatory, which can be obtained without any tense/aspect marker, as in (1). In fact, the markers are even forbidden there, as opposed to V O *de* clefts where they are allowed:

(4) a. Tense/aspect marker in V *de* O

*Shi wo dapo (*le) de na-ge huaping*⁵.
be 1SG break PFV DE that-CL vase
'It was I who broke that vase.'

b. Tense/aspect marker in V O *de*

Shi wo dapo (le) na-ge huaping de.
be 1SG break PFV that-CL vase DE
'It was I who broke that vase.'

Focusing on the V *de* O pattern in (northern) Mandarin Chinese⁶, this paper aims to answer

³ In addition to the patterns shown here, bare *shi/de* sentences with similar meanings have been considered further subtypes of Chinese clefts (often under the same cover term *shi...de construction*). However, the bare patterns will not be discussed in this paper: bare-*shi* seems to involve a different structure, while bare-*de* cases can mostly be seen as variants of the non-bare version with the copula dropped (indeed, all examples presented here can retain their acceptability/interpretation with *shi* removed).

⁴ Descriptive studies often depict the variation as the sentence-final particle *de* optionally preceding the object (e.g., Li, Thompson, and Zhang 1998: 95-96; Yuan 2003: 6).

⁵ Note that this should be distinguished from the right dislocation case, which does allow the perfective marker *le*: right dislocation comes with a distinct prosodic pattern, with a potential pause preceding the right-dislocated object. In that case, the pattern involved would be V O *de* instead of V *de* O.

⁶ As is well noted, the V *de* O order is mainly restricted to northern Mandarin dialects, V O *de* being the predominant order in southern China (Simpson and Wu 2002: 169; Paul and Whitman 2008: 427-428). This paper focuses on the former in northern Mandarin, where both patterns are extensively present, while making comparisons where relevant.

the following questions: Is it a real equivalent to English *it*-clefts? In what respects are they similar/different? More specifically, is there really an overt focus-presupposition bipartition as often stated? If not, how is focus encoded? Perhaps even more importantly, how do we account for the peculiarities of V *de* O clefts as seen above?

Starting with common generalizations about this pattern, I show that not all of them hold when subjected to closer scrutiny: Some do (e.g., exhaustive focus, obligatory past time reading), while some do not (e.g., strict adjacency effect, term focus restriction). Although an overt focus-presupposition bipartition is shown to be absent, I agree with previous scholars in attributing the focus interpretation to *shi*: specifically, it selects a FocP as its complement (cf. Belletti’s 2009, 2015 analysis for English/French-type clefts). The peculiarities of the V *de* O pattern, on the other hand, can be explained if we take the functional element *de* to head an Asp*P encoding perfective aspect, which existentially closes off an event (Ramchand and Svenonius 2014). The V *de* O clefts are derived when an extended “eventive” projection minimally containing Asp*P is focus-moved to SpecFocP (often in a pied-piped way).

The paper is organized as follows. In Section 2, I introduce the common generalizations about V *de* O clefts before turning to the proposals built on them. Section 3 discusses the challenging data for previous accounts. Section 4 presents the new analysis, which not only accounts for the updated data range but also makes the right prediction on predicate restrictions (as shown in Section 5). Section 6 concludes with further issues/implications for future research.

2. Previous discussions

This section introduces previous observations and proposals about V *de* O clefts, focusing on three most influential analyses from the last two decades (Simpson and Wu 2002; Paul and Whitman 2008; Hole 2011).

2.1. Generalizations

The following properties are claimed to be characteristic of Chinese V *de* O clefts (Paul and Whitman 2008: 429–433; Hole 2011: 1710–1714):

- i. The sentence has to be interpreted with an exhaustive focus;
- ii. The focused element must be adjacent to the copula;
- iii. A past time reading of the presupposition is obligatory;
- iv. Markers of tense/modality/aspect(TMA) cannot occur in the presupposition part;
- v. Only arguments/adjuncts can bear the focus, i.e., term focus only.

The exhaustive focus is illustrated below, where the V *de* O sentence cannot be followed by an utterance with an additive effect:

(5) Exhaustive focus

Shi wo mai de mianbao. (#Lusi ye mai le.)

be 1SG buy DE bread Lusi also buy PFV
 ‘It was I who bought the bread. (#Lusi also bought some.)’

Indeed, one cannot felicitously suggest another bread-buyer after identifying *wo* ‘I’ as the only buyer. Such behavior resonates with the standard claim about English *it*-clefts (e.g., Chomsky 1977; Kiss 1998), confirming the reasonable association between them.

Another argument for matching these cross-linguistic clefts is the adjacency between the focused constituent and the copula. This is shown straightforwardly in all the examples so far (see also the discussion below related to the term focus restriction).

Properties (iii–v), on the other hand, distinguish V *de* O clefts from their English counterparts⁷. As already partly shown, the presupposition has to be interpreted as a past event (6) and no TMA marker can appear within it (7):

(6) Future-oriented temporal adverb (+ future-marking modal)

* *Shi wo mingtian (hui) mai de mianbao.*⁸
 be 1SG tomorrow will buy DE bread
 Intended: ‘It is I who will buy the bread tomorrow.’

(7) TMA markers

* *Shi wo {hui/ keyi} mai {le} de mianbao.*
 be 1SG will can buy PFV DE bread
 Intended: ‘It is I who {will/can/has buy/bought} the bread.’

These restrictions are clearly not inherent to clefts in general, as evidenced by the well-formed English translations.

In terms of focusable categories, it is often claimed that only term focus (i.e., focus on arguments/adjuncts) is available in the V *de* O pattern. In addition to the subject clefts we have seen so far, adjunct clefts as below are also commonly noted:

(8) Adjunct focus

Wo shi zai zheli mai de mianbao.
 1SG be at here buy DE bread
 ‘It was here that I bought the bread.’

Much less discussed is another type of term clefts, where the object is focused (therefore stressed, indicated by underline⁹):

⁷ In fact, properties (iii–v) also distinguish V *de* O clefts from the similar looking V O *de* pattern, but a comprehensive investigation into all patterns of Chinese clefts will have to wait for future research.

⁸ Note that this sentence can sound right with a different interpretation: ‘(It) is the (kind of) bread that Zhangsan will buy tomorrow.’ That would become a case of simple predication, with a dropped *pro* as subject and a noun phrase modified by a relative clause as predicate.

⁹ Focused elements are stressed by default, being phonologically more prominent than the non-focus parts in the sentence; it is explicitly marked here because the same sequence can yield different focus interpretations, each of them associated with a specific prosodic pattern (see Section 3.2).

(9) Object focus

Wo shi mai de mianbao.

1SG be buy DE bread

‘It was bread that I bought.’

Cases like this have been mostly ignored by previous studies on Chinese clefts: they clearly violate the adjacency restriction. Unlike what we have seen so far with the canonical (i.e, subject/adjunct) clefts, an object can get focused in a non-copula-adjacent position. As a matter of fact, the object simply cannot be fronted to the copula-adjacent position (10a), even though object fronting does exist elsewhere in Chinese (10b/c).

(10) a. Object fronting in V *de* O clefts

* *Wo shi mianbao mai de.*

1SG be bread buy DE

Intended: ‘It was bread that I bought.’

b. Object fronting in non-clefts I

Wo mianbao mai le (, niunai mei mai).

1SG bread buy PFV milk NEG.past buy

‘I bought bread (, but not milk).’

c. Object fronting in non-clefts II

Mianbao wo mai le (, niunai wo mei mai).

bread 1SG buy PFV milk 1SG NEG.past buy

‘Bread I bought (, milk I didn’t).’

In spite of their distributional deviance, V *de* O sentences with object focus still count as clefts, since they do behave on a par with the more typical types otherwise (Paul and Whitman 2008: 428–429, footnote 15). The contradiction between adjacency and object focus thus poses a challenge to any analysis aiming at descriptive adequacy; in the next subsection, we will see how previous research approach it.

2.2. *Proposals*

Despite the consensus on treating V *de* O sentences as a cleft construction, previous opinions vary considerably concerning their internal structure. The focus interpretation is mostly associated to *shi*: some straightforwardly label it as a focus marker (e.g., Yuan 2003), while most still see it as a copula or functional verbal/auxiliary element (e.g., Simpson and Wu 2002; Paul and Whitman 2008). For many, an overt focus-presupposition bipartition also exists in Chinese clefts, accounting for their similarity to English *it*-clefts (Paul and Whitman 2008; Hole 2011). The distinctive properties, on the other hand, are mostly attributed to *de*.

Previous descriptivists have treated it as a nominalizer (Zhu 1978; Paris 1979), a tense/aspect marker (Song 1981; Shi 1984), or a preposed sentence-final particle (Li, Thompson, and Zhang 1998; Yuan 2003), while more recent generative studies have analyzed it as T⁰ (Simpson and Wu 2002), Asp⁰ (Paul and Whitman 2008), or C⁰ (Hole 2011).

Let us take a closer look at the last three analyses to see how they address (or not address) the empirical generalizations observed:

Simpson and Wu (2002) are among the first to bring the V *de* O pattern into theoretical discussions. They observe the important distinction between different patterns of Chinese clefts in terms of possible time readings and suggest distinct treatments for past-only ones and non-past ones. More specifically, they claim that *de* in V *de* O clefts is a past tense morpheme heading a TP, which is selected by *shi*, a higher verbal/auxiliary element. The past time restriction is thus explained, along the same lines of Stowell's (1996) double-tense analysis for English present perfect. What they fail to mention is how the focus effects come about in their theory.

Paul and Whitman (2008) do take focus into account. They associate the exhaustive focus with adjacency and assume the presence of an overt focus-presupposition partition in Chinese clefts. As with English *it*-clefts, this bipartition is also allegedly selected by the copula; however, in the Chinese case, it is headed by an Asp⁰, which is lexicalized by the verb-adjacent *de*. The complement of the copula being smaller than TP naturally explains the ban on TMA markers, and a [past] tense feature on *de* is assumed to account for the obligatory past reading. Object clefts are left untreated due to the violation of adjacency, the latter playing a central role in their analysis: they take object focus to involve a different mechanism (Paul and Whitman 2008: 431, footnote 18).

Contrary to Paul and Whitman (2008), Hole (2011) calls for a unified treatment of all *shi...de* clefts, including those with sentence-final *de*. Adopting the bipartition view, he attempts to resolve the contradiction between adjacency and object clefts with a highly modular proposal: object clefts are claimed to be the only exception where the partitioning only takes place at LF. *De* is treated as a C head carrying an [exhaustive] feature, which attracts the fronted focus and leads to the bipartition. As for the temporal restriction and TMA ban, Hole (2011: 1714) assumes that the presupposition of V *de* O clefts does contain TMA information, with their default values being [+anterior, -irrealis, +terminative]; explicit TMA material is forbidden due to PF-level constraints. Finally, *shi* is seen as a predication copula, which simply mediates between topics/subjects and comments/predicates.

3. Challenging evidence

In this section, I present some evidence challenging two of the generalizations introduced in the last section: the adjacency restriction and the term focus restriction.

3.1. Focus shifting via stress placement

Contrary to the adjacency claim, the copula is not always immediately followed by the focus. In addition to clear cases with object focus as seen earlier, V *de* O clefts with a shiftable focus also demonstrate the lack of adjacency:

(11) Focus shifting via stress placement

Wo shi zuotian qi che qu de Faguo.

1SG be yesterday ride bike go DE France

i. ‘It was yesterday that I went to France by bike.’

ii. ?‘It was by bike that I went to France yesterday.’

iii. ?‘It was France that I went to by bike yesterday.’

iv. ‘I went to France BY BIKE YESTERDAY.’

v. *‘It is that I went to France by bike yesterday.’

According to the strict adjacency/bipartition view, (i) should be the only possible reading we can get, with the copula-adjacent adjunct *zuotian* ‘yesterday’ in focus and thus stressed. However, if we move the narrow stress to other constituents or simply remove it, the same sentence can be interpreted differently, as shown by the readings above: (ii/iii) can be obtained with stress assigned to the respective constituent in focus and (iv) is the reading we get if the sentence is uttered with a relatively neutral prosodic pattern (the last reading, however, is never possible; see the discussion below).

As can be observed, (ii) and (iii) are obtained with somewhat degraded naturalness¹⁰. It seems to result from some pragmatic factor: There simply exists another preferred way to express the same meaning, namely putting ‘yesterday’ before the copula, since non-focused information is usually given and thus tends to occur in a topical position.

The careful reader may have noticed the distinction in the translations for (i–iii) and for (iv). Indeed, for a non-contrastive/neutral context where reading (iv) is most readily available,

¹⁰ This is sometimes argued to save the validity of the adjacency restriction. Hole (2011: 1712), for instance, takes it to mean that only certain contexts can license the prosodically-marked focus, which differs from, and affects the prosodic behavior of, the copula-adjacent ‘cleft focus’. More specifically, he claims that readings like (ii) are only available in corrective situations like below, where ‘the cleft focus is downgraded prosodically due to its second-occurrence status’:

a. Non-adjacent focus in corrective context

A: *Tingshuo ni shi zuotian zuo huoche qu de Faguo.*

hear.say 2SG be yesterday sit train go DE France

A: ‘I heard that it was yesterday that you went to France by train.’

B: *Bu, wo shi zuotian QI CHE qu de Faguo.*

no 1SG be yesterday ride bike go DE France

B: ‘No, it was yesterday that I went to France BY BIKE.’

(adapted from Hole 2011: (14’))

This line of reasoning, however, still fails to explain readings like (iv), where ‘by bike yesterday’ cannot be treated as one clefted constituent (cf. Cheng 2008: 250 for discussion on the similar behavior of V O *de* pattern).

the in-situ focusing strategy seems preferred to clefting in English:

(12) Neutral context for reading (iv)

A: *Tingshuo ni qu Faguo le.*
hear.say 2SG go France SFP

A: 'I heard that you went to France.'

B: *Dui, wo shi zuotian qi che qu de (Faguo).*
right 1SG be yesterday ride bike go DE France

B: 'Yes, I went {there/to France} BY BIKE YESTERDAY.'

In such situations, the V *de* O sentence just serves to add new information about the presupposed event ("speaker B going to France" in this case), without contrasting with or correcting anything in the previous discourse, or the common ground.

However, the proposition focus reading (v) is never possible, even uttered with a neutral prosodic pattern in an appropriate context. Imagine that speaker B lives in a Swiss city bordering France and that crossing the border by bike is possible, though much more tiring than taking public transport. (11) still cannot be a felicitous answer to a *how-come* question like below:

(13) Neutral context for intended reading (v)

A: *Ni zenme kanqilai zheme lei?*
2SG how.come look so tired

A: 'How come you look so tired?'

B: # *Wo shi zuotian qi che qu de Faguo.*¹¹
1SG be yesterday ride bike go DE France

B: '(It is that) I went to France by bike yesterday.' (intended)

The availability of (iv) and unavailability of (v) shows an interesting contrast: it seems to suggest that the event in question has to be *existentially presupposed*, considering that its existence cannot be conveyed as new information.

¹¹ Dropping the copula *shi* appears to rescue the felicity of the answer, but the focal stress would then necessarily fall on *qi che* "by bike", forcing the rest of the sentence to be presupposed and yielding a narrow focus interpretation: "It was by bike that I went to France yesterday."

The proper way to express proposition focus in this case would be to use a non-cleft sentence with a sentence-final particle *le* (see Soh 2009 and references therein for detailed discussions on this particle):

- a. *Wo zuotian qi che qu Faguo le.*
1SG yesterday ride bike go France SFP
'(It is that) I went to France by bike yesterday.'

3.2. More focusable categories

As mentioned earlier, focus in V *de* O clefts is often claimed to fall on arguments/adjuncts only. A closer examination proves otherwise: With the right predicate, prosody and context, V *de* O clefts can also express non-term focus, as shown in (d–f) below (term focus examples repeated in (a–c); focus indicated by underline).

(14) a. Subject focus

Shi *wo* *mai* *de* *mianbao*.

be 1SG buy DE bread

‘It was I who bought the bread.’

b. Adjunct focus

Wo *shi* *zai* *zheli* *mai* *de* *mianbao*.

1SG be at here buy DE bread

‘It was here that I bought the bread.’

c. Object focus

Wo *shi* *mai* *de* *mianbao*.

1SG be buy DE bread

‘It was bread that I bought.’

d. Verb focus

Wo *shi* *mai* *de* *mianbao*.

1SG be buy DE bread

‘I BOUGHT the bread. (I didn’t BAKE it myself.)’

e. VP focus

Wo *shi* *mai* *de* *mianbao*. (*Lusi* *shi* *shua* *de* *wan*.)

1SG be buy DE bread Lusi be brush DE bowl

‘I bought the bread. (Lusi did the dishes.)’

[as an answer to “What did you and Lusi do for the family dinner?”]

f. Proposition focus

Shi *wo* *jiao* *zhu* *de* *ta*.¹²

be 1SG call stop DE 3SG

‘(It is that) I stopped her/him from leaving (by calling).’

[as an answer to “Why is (s)he still here?”]

(Li, Thompson, and Zhang 1998, (4’))

As can be observed, all examples above show the temporal restriction: None of them allow

¹² This sentence can of course have a narrow focus reading, too, with the subject alone in focus.

a non-past reading. The ban on TMA markers seems universal, too: Their appearance would render (d–f) ungrammatical, just as it would do (a–c) (cf. (7) from Section 2.1). In terms of exhaustivity, the translations of (d–f) may not serve as transparent indications (unlike those of (a–c)), but this does not mean that non-term focus in V *de* O clefts can be non-exhaustive. In fact, the effect can be better detected once we consider another way of translating (d–f):

(14) d'. Verb focus

‘I BOUGHT the bread. (I didn’t BAKE it myself.)’ OR:
 ‘It is buying that I did to get the bread (, not baking).’

e'. VP focus

‘I bought the bread.’ OR:
 ‘It was buying the bread that I did (for the family dinner).’

f'. Proposition focus

‘(It is that) I stopped her/him from leaving (by calling).’ OR:
 ‘It was my stopping her/him from leaving (that led to her/him still being here).’

By rephrasing the translations in the form of *it*-clefts, we can now capture the exhaustive effect with non-term focus as well as with term focus.

Another crucial observation can be made on the basis of these rephrased translations: compared to term focus, the expression of non-term focus with V *de* O clefts appears more selective in terms of context and predicate type. For instance, the translation in (14d') shows that to get a verb focus cleft, the result state entailed by the verb has to be existentially presupposed (i.e., “I already have/own the bread” here) and there should be alternative ways to achieve this state. (14e') also demonstrates the importance of an appropriate context.

The proposition focus in (14f), compared with the impossible case in (13) above (repeated below with context omitted), reveals further restrictions:

(15) a. Proposition focus (= 14f)

Shi wo jiao zhu de ta.
 be 1SG call stop DE 3SG
 ‘(It is that) I stopped her/him from leaving (by calling).’

b. Intended proposition focus in (13)

* *Wo shi zuotian qi che qu de Faguo.*
 1SG be yesterday ride bike go DE France
 Intended: ‘(It is that) I went to France by bike yesterday.’

A noticeable difference between the above two cases lies in the position of the subject: it follows the copula in (15a) but precedes the copula in (15b). However, the latter is not ruled out simply for distributional reasons, as the proposition focus reading remains inaccessible

with a post-copula subject:

- (16) Intended proposition focus with post-copula subject
* *Shi wo zuotian qi che qu de Faguo.*
be 1SG yesterday ride bike go DE France
Intended: ‘(It is that) I went to France by bike yesterday.’

Here again, the translations may give us a clue. If we try to rephrase the translation for (13/15b) in the same way as we did for (14f/15a), we have the following pair:

- (15) a'. Proposition focus
‘(It is that) I stopped her/him from leaving (by calling).’ OR:
‘It was my stopping her/him from leaving (that led to her/him still being here).’

b'. Intended proposition focus
Intended: ‘(It is that) I went to France by bike yesterday.’ OR:
Intended: ‘It was my going to France by bike yesterday *(that led to me looking tired).’

The noticeable difference here concerns the (im)possibility of dropping the presupposed result state in the translation. While (15a') allows the result state to be dropped as a presupposed entailment, (15b') does not. What really rules out (15b/16) thus seems to be related to predicate type: “Stopping her/him from leaving” naturally entails “her/him still being here” as the result state, which can be dropped if presupposed, whereas “going to France by bike” does not necessarily lead to “looking tired” the following day¹³, hence the impossibility to drop the presupposition. In other words, a causal relation *per se* does not warrant the availability of proposition focus with V *de* O clefts; it has to be weaved into the denotation of the predicate.

In view of this point, the earlier conjecture needs to be revised as follows: The *whole event* is not necessarily existentially presupposed; the *result state* entailed by it is (which of course means, or presupposes, that the event must have an inherent result state).

To the best of my knowledge, most of the new data discussed here remain untouched, thus unpredicted, by previous generative work on V *de* O clefts (see Yang and He 2021 for a recent constructionist view, though). In the following, I will propose a new analysis which not only takes into account all the data available, but also makes the right prediction.

4. A cartographic analysis

With a more complete empirical picture in mind, we can now attempt to answer the questions raised at the beginning of this paper:

¹³ Depending on how far your starting point is from the border, how good a cyclist you are and/or how well you have slept afterwards.

- Is the Chinese *V de O* pattern a real equivalent to English *it*-clefts? In what respects are they similar/different?

Answer: In a way yes, since all Chinese *V de O* clefts can be paired with an English translation in the form of *it*-clefts (albeit less natural in some cases), as we have seen in the last section. This parallel demonstrates their common association with exhaustive focus.

But the former is remarkably more restricted. As pointed out by previous work, no TMA markers are allowed in the presupposition part and a past reading is obligatory in *V de O* clefts, whereas there are no such constraints for *it*-clefts. In addition, *V de O* clefts show higher selectivity in terms of context, predicate type, etc.

- Is there an overt focus-presupposition bipartition?

Answer: No. The sentence cannot be bipartitioned when the focus is not adjacent to the copula, e.g., on the object or the second adjunct.

- How is focus encoded?
- How do we account for the peculiarities of *V de O* clefts?

Answers: I follow most existing literature in attributing the focus interpretation to *shi* and the peculiarities of *V de O* clefts to *de*. More specifically, I believe that each of these two elements is associated with a key functional projection: *shi* with FocP and *de* with Asp*P (related in different ways). Below I discuss how exactly focus is encoded in the functional hierarchy revolving *shi* (Section 4.1) and how the syntactic status of *de* helps to account for the particular behaviors observed of this pattern (Section 4.2).

4.1. Copula *shi* and FocP

Following Paul and Whitman (2008) and Hole (2011), I consider *shi* in Chinese *V de O* clefts to be a copular verb, instead of a focus marker. This can be easily captured by examples like below:

(17) Negation + *shi*

Wo bu shi zai zheli mai de mianbao.
 1SG NEG be at here buy DE bread
 ‘It was not here that I bought the bread.’

Given that it can be negated with the sentential negation marker *bu*, *shi* must be a verbal/predicative element, instead of a fully grammaticalized focus marker (as with English *it*-clefts). But it is arguably not far from focus markers if we adopt the cartographic approach (Rizzi 1997; Cinque and Rizzi 2010; Rizzi and Cinque 2016).

Rizzi (1997) proposes to decompose the CP layer (i.e., the left periphery) into a sequence of hierarchically organized functional heads:

(18) ForceP > TopP* > FocP > TopP* > FinP > IP (Rizzi 1997: 279)¹⁴

¹⁴ The asterisk indicates that Italian topics can be reiterated.

Among these projections, FocP is claimed to host focus information in its specifier, including fronted *wh*-phrases in languages with *wh*-movement. Crucially, the head of FocP can be lexicalized by focus markers in languages which have them (see, e.g., Aboh 2004 for Gungbe; cf. Durrleman and Shlonsky 2015 for Jamaican Creole).

Now if we return to clefts, recent cartographic studies (Belletti 2009, 2015; Haegeman, Meinunger, and Vercauteren 2015) have claimed a close relationship between the copula and the FocP (its head being null) for English-type clefts¹⁵. More specifically, a left peripheral FocP is selected by the copula as its complement. This “embedded” analysis is shown to fare better than the “matrix” accounts (Meinunger 1998; Frascarelli and Ramaglia 2013) in terms of predictions (see Haegeman, Meinunger, and Vercauteren 2015 for more detailed arguments).

To capture the crosslinguistic parallel, I assume that a similar functional hierarchy also exists in Chinese *V de O* clefts: *shi* does not lexicalize Foc⁰ but stays a copular verb which selects FocP¹⁶ (with a null head) as its complement.

But what about the differences? Questions naturally arise when we consider the distinctive properties of the *V de O* pattern, notably the lack of an overt focus-background bipartition (which is at the origin of the name *clefts*). I argue that focus is indeed syntactically encoded there with a FocP and that there is also overt focus movement; but crucially, what is moved to SpecFocP differs from English-type clefts, accounting for the remarkable distinctions.

4.2. Nature of *de*: *Asp**

So, what exactly is focus-moved in Chinese *V de O* clefts? A simple and intuitive answer is the whole sequence “*V de O*”: The FocP hypothesis can be maintained as long as we assume this. A more revealing answer relies on the syntactic status of *de*. Earlier discussions around the rephrasable translations already reveal the essence of the proposal: FocP could host in its

¹⁵ Belletti (2009, 2015) observes a subject-object asymmetry in relation to different types of cleft focus: subject clefts can express either new information or corrective/contrastive focus, while object/non-subject clefts can only express the latter. Two Focus projections are employed to account for this: a left peripheral FocP dedicated to contrastive/corrective focus and a vP peripheral FocP (Belletti 2004) for new information focus.

¹⁶ Chinese *V de O* clefts do not seem to show the same asymmetry as mentioned in the last footnote. For instance, all examples in Section 3.2 with different constituents in focus can be used as felicitous answers to *wh*-questions, thus expressing new information focus (as shown by the explicit contexts in Section 3 and exemplified below by an object cleft).

- | | | | | | | | | | | | |
|----|----|---------------------|------------|-----------------|----------------|----|-------------------|--------------|------------|-----------|-----------------|
| a. | A: | <i>Ni</i> | <i>mai</i> | <i>{le/ de}</i> | <i>shenme?</i> | B: | <i>(Wo)</i> | <i>(shi)</i> | <i>mai</i> | <i>de</i> | <i>mianbao.</i> |
| | | 1SG | buy | PFV | DE | | 1SG | be | buy | DE | bread |
| | | ‘What did you buy?’ | | | | | ‘I bought BREAD.’ | | | | |

It appears that at least in the Chinese pattern investigated here, the type of focus involved is invariably identificational/exhaustive (Kiss 1998; Cruschina 2021 among others), with further subtypes (new information vs contrastive/corrective) derivable in appropriate contexts. I thus leave open which FocP is involved here while keeping the core insight of the embedded analysis: the copula in clefts necessarily selects a FocP as its complement (with possible crosslinguistic variation in the specific type of focus hosted). I thank the anonymous reviewer who asked for clarification on this point.

specifier some “eventive” projection, containing minimally the verb and the object. The functional element in between thus most likely belongs to the extended verbal domain.

Here I propose that *de* heads some sort of Asp*P, an aspectual projection that specifically encodes perfective aspect and existentially closes off an event (Ramchand and Svenonius 2014). Assuming the existential closure of the event has many merits. To begin with, it immediately explains away the necessary past-time/anterior reading, which comes about as soon as the perfective aspect is combined with the speech time. TMA markers are banned simply because their denotations contradict what is already encoded by Asp*P: an existentially closed event cannot be further quantified/split/closed again. Another advantage is that it corresponds nicely to the observation that only events entailing an inherent result state can be clefted with a proposition focus reading (cf. Section 3.2), as the perfective aspect intrinsically requires a result state too. Syntactically speaking, a projection smaller than Asp*P does not qualify to be focus-moved to SpecFocP.

Now that we have the key ingredients outlined, the movement to derive V *de* O clefts can be schematized as below:

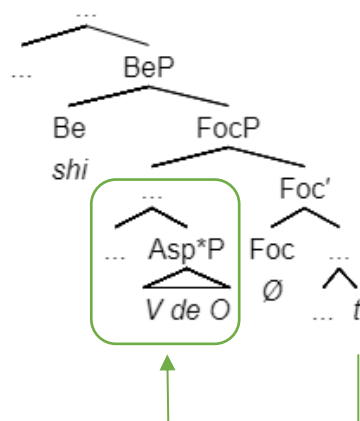


Figure 1 Overt focus movement in Chinese V de O clefts

In brief, V *de* O clefts are derived when an extended “eventive” projection containing Asp*P is focus-moved to SpecFocP. Crucially, the focus feature triggering the movement can be carried by different elements within the moved projection, accounting for the various focusable categories. In a minimal case where only Asp*P is moved, either the verb or the object, or both together, are focused semantically as (part of) the event description. When the subject/adjunct bears the focus, it pied-pipes its containing projection (larger than Asp*P) as an event modifier. Finally, as a maximal case, an event as a whole can be focused, with its entailed result state presupposed.

5. Prediction: predicate restrictions

Once we have a plausible analysis that can account for all the available data, it should also be able to make testable predictions. One of such predictions made by the new analysis

concerns predicate types.

As mentioned earlier, the cleft patterns *V de O* and *V O de* have been assumed as equivalents by some, due to their apparent interchangeability in certain contexts. However, if we take the verb-adjacent *de* to be a perfectivity-encoding aspectual head, distinct from the clause-final *de*, the two patterns must behave quite differently in terms of the compatibility with predicates of different lexical aspect, or *Aktionsart*.

Starting with the *V de O* pattern, it is expected to show intolerance only to stative predicates, given their difficulty to combine with the perfective aspect (regardless of which element is focused). This seems to be the case indeed:

(19) *V de O* + different predicate types

*Wo shi {*xihuan/ mai/ dapo} de na-ge huaping.*
 1SG be like buy break DE that-CL vase
 ‘It is/was that vase that I {*like/bought/broke}.’

As for the *V O de* pattern, the situation turns out to be more complicated, showing variation according to the focused constituent. For example, only telic verbs are allowed with subject focus:

(20) Subject focus + different predicate types

*Shi wo {*xihuan/ *mai/ dapo} na-ge huaping de.*
 be 1SG like buy break that-CL vase DE
 ‘It is/was I who {*likes/*bought/broke} that vase.’

Adjunct focus allows both telic verbs and statives:

(21) a. Adjunct focus + stative predicate

Wo shi yinwei zhe-ge yuanyin xihuan na-ge huaping de.
 1SG be because.of this-CL reason like that-CL vase DE
 ‘It is for this reason that I like that vase.’

b. Adjunct focus + eventive predicates

*Wo shi zai zheli {?*mai/ dapo} na-ge huaping de.*
 1SG be at here buy break that-CL vase DE
 ‘It was here that I {bought/broke} that vase.’

Object focus demonstrates yet another pattern: only statives (marginally) allowed.

(22) a. Object focus + stative predicate

? *Wo shi xihuan na-ge huaping de.*
 1SG be like that-CL vase DE
 ‘It is that vase that I like.’

b. Object focus + eventive predicates

*Wo shi {*mai/ *dapo} na-ge huaping de.*

1SG be buy break that-CL vase DE

‘It was that vase that I {bought/broke}.’

The varied interaction between different patterns, focus types and predicate types can be summarized as below:

Table 1 Varied predicate restrictions in Chinese shi...de clefts

Pattern & focus type	Stative (e.g., <i>like</i>)	Eventive	
		Atelic (e.g., <i>buy</i>)	Telic (e.g., <i>break</i>)
V <i>de</i> O (all types)	*	OK	OK
Subject focus	*	*	OK
V O <i>de</i> Adjunct focus	OK	*	OK
Object focus	?OK	*	*

Interestingly, every two adjacent rows in the table overlap without being identical. This puzzle, if not entirely a coincidence, deserves some principled explanation. A desirable theory of Chinese *shi...de* clefts should be able to capture both the distinctions and the similarities reported. For our purpose here, it suffices to observe the distinctive results for the two patterns as predicted.

6. Conclusion

In this paper I have investigated the syntax and semantics of Chinese V *de* O clefts. I have shown that while some of the common generalizations hold upon closer examination (e.g., exhaustive focus, obligatory past time reading), some do not (e.g., strict adjacency effect, term focus restriction). I have argued for the existence of two key functional projections, each linked to one crucial element in the pattern: (i) a FocP selected by the copula *shi*; (ii) an Asp*P headed by *de*. The former explains the focus effects with syntactic encoding, while the latter accounts for the temporal/aspectual peculiarities by marking perfective aspect (related predictions on predicate restrictions also presented). The derivation of V *de* O clefts involves overt focus movement of an extended “eventive” projection, which minimally contains Asp*P. This movement is triggered by a focus feature either on the whole constituent or on a smaller constituent within it (the latter involving pied-piping).

If this approach proves to be on the right track, many more interesting questions arise. Below are some of the topics/implications to be explored in future research:

What are the omitted derivational steps to complement the creation of V *de* O clefts? If *de* is a perfective marker, what distinguishes it from the verbal *le* (the widely accepted perfective

marker in canonical sentences)?

How do V O *de* clefts differ from and relate to V *de* O? Is there any common nature between the verbal *de* and the sentence-final *de*? Is a unified analysis possible for all Chinese cleft patterns, bare or non-bare (cf. Cheng 2008)?

Assuming the similar function of the copula, what then underlies the different “clefting” strategies in Chinese and English? What parallels can be drawn between *de* and *-en*, the respective Asp* heads in the two languages (cf. Simpson and Wu 2002)?

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